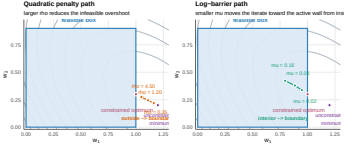
# Optimization in Machine Learning

## Constrained Optimization
## Penalty-Based and Primal-Dual Methods




### Learning goals

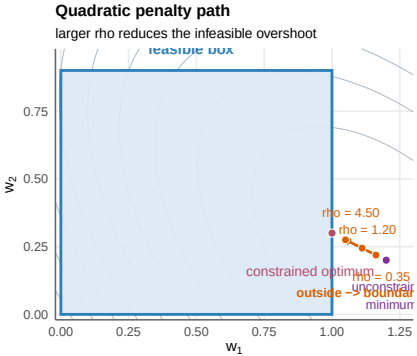
- Quadratic penalties for violated constraints
- Log-barrier and interior-point ideas
- Why penalty / barrier parameters must change over time
- Perturbed KKT systems and primal-dual interior-point updates

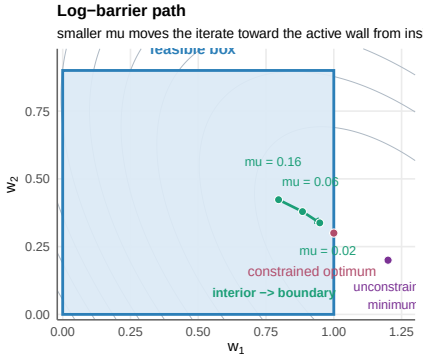
# FROM DUALITY TO HYBRID METHODS

**Starting point:**

$$\min_{\boldsymbol{x}\in\mathbb{R}^d} F(\boldsymbol{x}) \qquad \text{s.t.} \qquad g_i(\boldsymbol{x}) \le 0,\ h_j(\boldsymbol{x}) = 0.$$

- Projection kept iterates feasible; duality relaxed constraints through multipliers.
- **Penalty methods** allow violations but charge them inside the objective.
- **Barrier methods** stay strictly feasible and approach the boundary from the interior.



# PENALTY PATH ON THE BOX TOY

Reuse

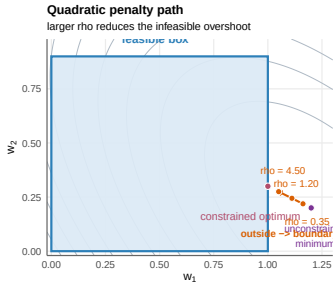
$$\min_{\boldsymbol{w}} F(\boldsymbol{w}) = (w_1 - 1.2)^2 + (w_1 - 1.2)(w_2 - 0.2) + (w_2 - 0.2)^2$$

$$\text{s.t.} \qquad 0 \le w_1 \le 1, \qquad 0 \le w_2 \le 0.9.$$

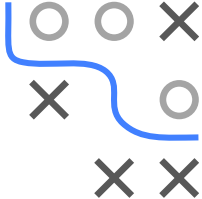
Quadratic penalty relaxation:

$$R(\boldsymbol{w}, \rho) := F(\boldsymbol{w}) + \frac{\rho}{2} \sum_{i=1}^{4} \max\{g_i(\boldsymbol{w}), 0\}^2.$$

- The unconstrained minimizer $(1.2, 0.2)$ violates only the upper bound $w_1 \le 1$.
- For moderate $\rho$, the relaxed minimizer stays slightly outside the box.
- As $\rho$ grows, the penalty path approaches the constrained optimum $(1, 0.3)$ from the infeasible side.

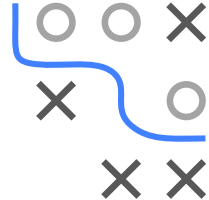


# EXACT VERSUS APPROXIMATE INNER SOLVES



- In penalty methods, feasible points receive **no extra cost** beyond the original objective.
- Exact statement: if the exact minimizer of the relaxed problem is feasible for the original constraints, then it also solves the original constrained problem.
- In practice, each relaxed problem is solved only approximately.
- Therefore, feasibility of the returned inner iterate by itself is **not** a proof of optimality for the original constrained problem.
- A safe continuation strategy is:
  1. choose an initial penalty weight $\rho_0 > 0$
  2. minimize $R(\boldsymbol{w}, \rho_t)$ to sufficient inner accuracy
  3. if the returned point is feasible and the inner solve is accurate enough, we may stop
  4. otherwise increase $\rho_t$ and solve again
- A common choice is geometric growth, for example $\rho_{t+1} = 2\rho_t$.
- The solution from the previous outer step is a natural warm start for the next relaxed problem.

# GENERAL QUADRATIC PENALTY RELAXATION



- Consider the constrained problem

$$\min_{\boldsymbol{x}} F(\boldsymbol{x}) \qquad \text{s.t.} \qquad g_i(\boldsymbol{x}) \leq 0,\ i = 1, \ldots, m, \qquad h_j(\boldsymbol{x}) = 0,\ j = 1, \ldots, k.$$

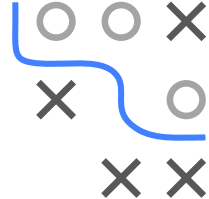
- A standard quadratic penalty relaxation is

$$R(\boldsymbol{x}, \rho) := F(\boldsymbol{x}) + \frac{\rho}{2} \left( \sum_{i=1}^{m} \max\{0, g_i(\boldsymbol{x})\}^2 + \sum_{j=1}^{k} h_j(\boldsymbol{x})^2 \right), \qquad \rho > 0.$$

- Inequality constraints are penalized only when violated; equality constraints are penalized whenever they are nonzero.
- Its gradient is

$$\nabla_{\boldsymbol{x}} R(\boldsymbol{x}, \rho) = \nabla F(\boldsymbol{x}) + \rho \sum_{i=1}^{m} \max\{g_i(\boldsymbol{x}), 0\} \nabla g_i(\boldsymbol{x}) + \rho \sum_{j=1}^{k} h_j(\boldsymbol{x}) \nabla h_j(\boldsymbol{x}).$$

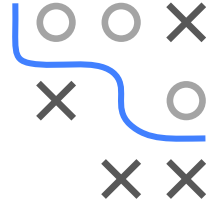
# PENALTY METHODS: PRACTICAL BEHAVIOR



- Penalty relaxations are attractive because they let us reuse unconstrained solvers such as GD, coordinate descent, or Newton-type methods.
- However, the iterates of the relaxed problem need **not** be feasible while $\rho$ is still moderate.
- Choosing an enormous penalty weight from the start is usually a bad idea:
  - the penalty term dominates the geometry
  - the relaxed objective becomes ill-conditioned
  - descent methods can bounce or converge very slowly
- Therefore, one usually starts with a smaller $\rho$, solves a reasonably conditioned relaxed problem, and then increases $\rho$ gradually.
- Exact feasibility is often only approached in later outer iterations, once the penalty weight is large enough.

# BARRIER PATH ON THE BOX TOY

Barrier methods keep the inequality constraints explicit and replace the boundary by an infinite wall.



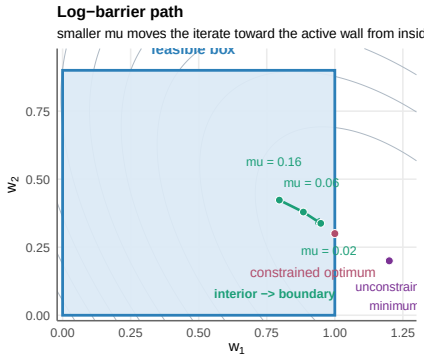
- For the same box toy,

$$B(\boldsymbol{w}, \mu) := F(\boldsymbol{w}) - \mu\Big(\log w_1 + \log(1 - w_1) + \log w_2 + \log(0.9 - w_2)\Big).$$

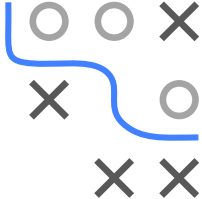
- It is defined only for **strictly feasible** points:

$$0 < w_1 < 1, \qquad 0 < w_2 < 0.9.$$

- As $\mu \downarrow 0$, the barrier path approaches $(1, 0.3)$ from the interior.



# BARRIER GRADIENT AND CONTINUATION



- For the logarithmic barrier, the gradient is

$$\nabla_{\boldsymbol{x}} B(\boldsymbol{x}, \mu) = \nabla F(\boldsymbol{x}) + \mu \sum_{i=1}^{m} \frac{\nabla g_i(\boldsymbol{x})}{-g_i(\boldsymbol{x})}.$$

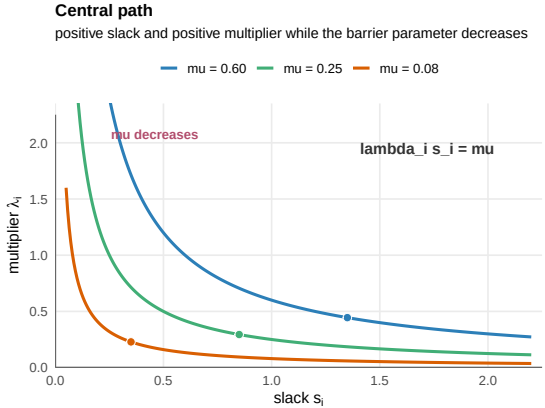
- As soon as one approaches the boundary $g_i(\boldsymbol{x}) \to 0^-$, the term $\nabla g_i(\boldsymbol{x})/(-g_i(\boldsymbol{x}))$ becomes very large.
- This is the barrier effect: the objective rises sharply near the boundary and prevents infeasible steps.
- The continuation strategy is the reverse of penalty methods:
  - start with a relatively large barrier weight $\mu$
  - solve the barrier problem
  - reduce $\mu$ and warm-start the next solve
- Large $\mu$ keeps the iterate well inside the feasible region; small $\mu$ allows it to move closer to the true constrained optimum.

# BARRIER STATIONARITY AND THE CENTRAL PATH

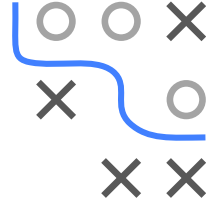
With slacks $s_i := -g_i(\boldsymbol{x}) > 0$ and multipliers $\lambda_i := \mu / s_i > 0$, barrier stationarity becomes

$$\nabla F(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\boldsymbol{x}) = \boldsymbol{0}, \qquad \lambda_i s_i = \mu.$$

- This is complementary slackness perturbed away from zero.
- The **central path** keeps slack and multiplier both positive while $\mu > 0$.



# PRIMAL-DUAL INTERIOR-POINT VIEW



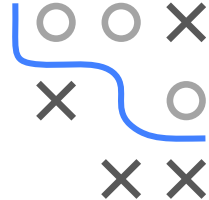
- Interior-point methods solve the primal and dual variables **together** through the perturbed KKT system

$$\nabla F(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\boldsymbol{x}) + \sum_{j=1}^{k} \nu_j \nabla h_j(\boldsymbol{x}) = \boldsymbol{0},$$

$$g_i(\boldsymbol{x}) + s_i = 0, \qquad h_j(\boldsymbol{x}) = 0, \qquad s_i > 0,\ \lambda_i > 0,$$
$$\lambda_i s_i = \mu, \qquad i = 1, \ldots, m.$$

- Newton steps are used to solve these equations approximately for a fixed $\mu$.
- Then $\mu$ is decreased and the next solve is warm-started from the current primal-dual pair.
- This is the clean primal-dual picture: the multipliers, slacks, and primal variables are coupled by one system instead of by an ad-hoc update rule.

# TAKE-AWAY



- **Penalty methods:** add violation costs and increase $\rho$ until feasibility is reached or closely approximated.
- **Barrier methods:** keep iterates strictly feasible and decrease $\mu$ to approach the optimum from the interior.
- **Primal-dual view:** barrier stationarity already contains multiplier information and leads directly to perturbed KKT conditions.
- **Trade-off:** penalties are simple but allow infeasible iterates; barriers preserve feasibility but require an interior start and careful continuation.